Fiscal Year:	FY 2021	**Task Last Updated:**	FY 01/01/2021
PI Name:	Matson, Douglas Ph.D.		
Project Title:	Electromagnetic Levitation Flight Support for Transient Observation of Nucleation Events		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	(617) 627-3239
Organization Name:	Tufts University		
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PI Web Page:	https://		
City:	Medford	**State:**	MA
Zip Code:	02155	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	12/01/2015	**End Date:**	02/29/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	4	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Sansoucie, Michael	**Contact Phone:**	256.544.5269
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: New grant number 80NSSC19K0256 awarded per NSSC information with period of performance of 3/1/2019-2/29/2024; this record for NNX16AB59G appended with that information per discussions with M. Sansoucie & L. Strutzenberg/MSFC (Ed., 6/21/19) NOTE: Extended to 11/30/2019 per NSSC information (Ed., 12/28/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC19K0256 ; NNX16AB59G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	EDITOR's NOTE (June 2019): New grant number 80NSSC19K0256 awarded per NSSC information with period of performance of 3/1/2019-2/29/2024; this record for NNX16AB59G (POP of 12/1/2015-2/28/2019) appended with that information per discussions with M. Sansoucie & L. Strutzenberg/Marshall Space Flight Center (MSFC). Continuation of "Electromagnetic Levitation Flight Support for Transient Observation of Nucleation Events" ; grant NNX14AB74G. See that project for previous reports. Funding is for Dr. Matson's role as Team Member on the subprojects described below, associated with European Space Agency (ESA)-sponsored main projects as noted. Main Project: Peritectic Alloy Rapid Solidification with Electromagnetic Convection (PARSEC); Science Coordinator is Dr. Thomas Volkmann, German Aerospace Center DLR, Institut für Materialphysik im Weltraum, Linder Höhe, Köln, Germany Subproject: Transformation kinetics - ISS (International Space Station) Using results from thermophysical property evaluation and combining these with the MHD convection definition, we will set convection over as wide a range as is possible and conduct recalescence studies from the undercooled melt using the Materials Science Laboratory-Electromagnetic Levitator (MSL-EML) to compare to results from ground-based EML and ESL. The critical parameters of interest are evaluation of the incubation time between formation of metastable phases and subsequent transformation to the stable phase for Fe-based and titanium aluminide alloys. Since this transformation is very rapid, the high-speed video system must be used to image the processes associated with nucleation and growth during phase selection. Main project: Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS); Science Coordinator is Professor Hans-Joerg Fecht, Ulm University, Ulm, Germany Subprojects: Management and Coordination. Coordination of the scientific program, Coordination of the ground based measurement program, Coordination of the microgravity experiment program, Contact with the national and international space agencies. Short Duration Microgravity Measurements - PF. Conduct parabolic flights (PF) on selected alloys. Long Duration Microgravity Experiments - ISS. Surface tension and viscosity as a function of temperature, Density as a function of temperature, Solidification growth velocity. Preparation of the Microgravity Experiments. Experimental protocols for processing on the ISS ; Verification of flight samples for the ISS. MODELLING A: Thermodynamics, Phase Selection. Thermodynamic phase diagram calculation, Modeling of the influence of convection on phase selection.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The value of this research is that the results have application to the design of industrial welding, spray forming, and casting operations for a commercially important class of soft magnetic materials. In addition, this research addresses fundamental issues relating to rapid solidification behavior, metastable phase selection, and analysis of the processes governing microstructural evolution. Experiments focus on the use of convection as a processing control parameter.
Task Progress:	Highlights of Work Accomplished During Grant Period Mar 2020 – Feb 2021. • Published nine peer-reviewed articles (see Bibliography section) and one Thesis. • Graduated two MS students (Azersky and Dytiuk). • Attended TMS (The Minerals, Metals & Materials Society)149th Annual Meeting, 23-27 February, 2020, in San Diego CA as Materials Science in Reduced Gravity symposium organizer and presenter • Attended American Society for Gravitational and Space Research (ASGSR)-36 2020 Annual Conference, November 4-9, 2020 virtual conference as national president, symposium organizer, and presenter • Featured speaker at Japan Microgravity Application Society Conference (JASMAC-32) and 181st Space Studies Board Meeting • European Space Agency 2020 Parabolic Fight Campaign, Inconel 625/L331 • PhD student Evan Baker recognized as ISS-National Lab Best Graduate Student Presentation at ASGSR-36 (2020) • PhD student Colby Azersky accepted to the NASA Pathways Intern Employment Program (2020) • Conducted flight planning for Batch 3 lead on samples FeCrNi 16-12 and FeCrNi 12-16, and CoSi eutectic in collaboration with the ESA LiPhas project • Successfully proposed Batch 4 testing with selected sample lead responsibility on Au/Pt, Inconel 625, and CMSX-4plus, and collaborations on High Entropy Alloys and FeCo samples. Highlights of Previous Work Accomplished. • Implemented Batch 1 and Batch 2 operations using ISS-EML on orbit. (Fe-20Cr-20Ni, FeCo50, FeCo55, TiZrNi, FeB, NdFeB, Ti48Al48Cr2Nb2, CMSX-10, LEK94, MC2); • Developed test plan documentation to run Batch 3/4 alloys using ISS-EML--(Fe-16Cr-12Ni, Fe-12Ni-16Cr, FeCo40, CoSi62); • Conducted parabolic flight and ground test experiment plan to support Batch 4 alloys (Au, CMSX-4+, EHEA,

	In625/L331, FeCo30); • Investigated thermophysical property measurement for control of convection using MSFC ESL and KRISS ESL in support of Batches 1-4; • Ran tests at Deutsches Elektronen-Synchrotron (English German Electron Synchrotron) synchrotron to define transformation pathway for alloy rapid solidification using Dresden EML facility and at APS Argonne National Lab using Tufts Glass-flux technique ; • Served as ESA Investigator Working Group (IWG) chair and NASA ISS-EML facility scientist over the entire program period; • Developed a new theoretical model to explain metastable transformations during rapid solidification of steel alloys to include the influence of melt convection with application to the welding and continuous casting industry ; • Developed a novel interpretation of how dislocation density plays an important role in defining how heterogeneous nucleation occurs along pre-existing grain boundaries in steel; impacts how investment casting systems are designed; • Developed a new approach to understanding the influence of deformation on the accuracy and precision of surface tension measurements for superalloys important for NASA exploration mission success; • Featured speaker for ISSR&D, ASGSR, ICASP-5 JASMAC-31, and JASMAC-32 conferences.
Bibliography Type:	Description: (Last Updated: 05/05/2026)
Articles in Peer-reviewed Journals	Xiao X, Rodriguez JE, Kelton KF, Matson DM. "Statistical learning for evaluation of crystal growth in low-melting alloy droplets with application to quasicrystal-forming Ti-Zr-Ni alloys." Modelling and Simulation in Materials Science and Engineering. 2020 Oct;28(8):085008, 1-22. https://doi.org/10.1088/1361-651X/abbfba , Oct-2020
Articles in Peer-reviewed Journals	Matson DM. "Influence of induced convection on transformation kinetics during rapid solidification of steel alloys: the retained damage model." JOM. 2020 Nov;72(11):4109-16. https://doi.org/10.1007/s11837-020-04375-2 , Nov-2020
Articles in Peer-reviewed Journals	Nawer J, Gosse S, Matson DM. "Tracking evaporation during levitation processing of nickel-based superalloys on the ISS." JOM. 2020 Sep;72(9):3132-9. https://doi.org/10.1007/s11837-020-04256-8 , Sep-2020
Articles in Peer-reviewed Journals	Zhao R, Wang Y, Gao J, Baker EB, Matson DM, Kolbe M, Chang AC, Ren Y. "In-situ and ex-situ studies of anomalous eutectic formation in undercooled Ni-Sn alloys." Acta Materialia. 2020 Sep 15;197:198-211. https://doi.org/10.1016/j.actamat.2020.07.048 , Sep-2020
Articles in Peer-reviewed Journals	Matson DM. "Maximizing scientific return from shared facilities working in space: ESA's investigator working group for the ISS-EML facility." Int J Microgravity Sci Appl. 2020;37(3):370301, 1-4. https://www.jstage.jst.go.jp/article/ijmsa/37/3/37_370301/_pdf , Jul-2020
Articles in Peer-reviewed Journals	Baker EB, Nawer J, Xiao X, Matson DM. "MHD surrogate model for control of convection in electromagnetically levitated molten metal droplets processed using the ISS-EML facility." NPJ Microgravity. 2020 Mar 16;6:9. https://doi.org/10.1038/s41526-020-0099-7 ; PMID: 32195320; PMCID: PMC7076000 , Mar-2020
Articles in Peer-reviewed Journals	Jeon S, Sansoucie MP, Shuleshova O, Kaban I, Matson DM. "Density, excess volume, and structure of Fe-Cr-Ni melts." J Chem Phys. 2020 Mar;152(9):094501, 1-7. https://doi.org/10.1063/1.5140787 , Mar-2020
Articles in Peer-reviewed Journals	Nawer J, Xiao X, Sancoucie MP, Matson DM. "Effect of mass evaporation on measurement of liquid density of Ni-based superalloys using ground and space levitation techniques." High Temperature High Pressures. 2020;49(1-2):17-29. https://doi.org/10.32908/hthp.v49.839 , Jan-2020
Articles in Peer-reviewed Journals	Bracker GP, Baker EB, Nawer J, Sellers ME, Gangopadhyay AK, Kelton KF, Xiao X, Lee J, Reinartz M, Burgraf S, Herlach D, Rettenmayr M, Matson DM, Hyers RW. "The effect of flow regime on surface oscillations during electromagnetic levitation experiments." High Temperature High Pressures. 2020;49(1-2):49-60. https://doi.org/10.32908/hthp.v49.817 , Jan-2020
Dissertations and Theses	Azersky C. "Evaluation of particle size dependent crystal defects in Inconel 718 by line profile analysis of x-ray diffraction." MS thesis, Mechanical Engineering Department, Tufts University, Medford MA 02155, August 2020. , Aug-2020